Calcurion Maths Worksheets

GCSE Mathematics

Topic: Ratio & Proportion

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Simplify the ratio 12 : 18. **[2]**

2. Simplify the ratio 24 : 36 : 48. **[2]**

3. Share £60 in the ratio 2 : 3. **[3]**

4. Share £120 in the ratio 1 : 2 : 3. **[3]**

5. Share £90 in the ratio 4 : 5. **[3]**

6. A recipe for 4 people uses 200 g of flour. How much is needed for 6 people? **[2]**

7. A recipe for 6 people uses 300 ml of milk. How much is needed for 10 people? **[3]**

8. 3 pens cost £1.80. How much do 5 pens cost? **[2]**

9. 5 kg of potatoes cost £3.50. How much do 8 kg cost? **[3]**

10. Write the ratio 0.5 : 1.5 in its simplest form. **[2]**

11. Express the ratio 2.5 : 7.5 in its simplest form. **[2]**

12. Divide 84 in the ratio 3 : 4. **[3]**

13. The ratio of boys to girls in a class is 3 : 5. There are 24 girls. How many boys are there? **[3]**

14. A map has a scale of 1 : 50000. What actual distance is represented by 4 cm on the map? **[3]**

15. A map has a scale of 1 : 25000. What length on the map represents 5 km? **[3]**

Section B – Hard (Q16–25)

16. Share £480 in the ratio 2 : 3 : 5. **[4]**

17. The ratio of red to blue counters is 3 : 7. There are 40 counters in total. How many are red? **[3]**

18. A mixture contains sand and cement in the ratio 5 : 2. How much cement is needed for 35 kg of mixture? **[3]**

19. 4 workers take 6 days to complete a job. How long would 8 workers take? **[3]**

20. 3 machines take 12 hours to produce 90 items. How long would 5 machines take to produce 150 items? **[5]**

21. A car travels 240 km on 30 litres of petrol. How far does it travel on 45 litres? **[3]**

22. The ratio $a : b = 4 : 5$ and $b : c = 3 : 2$. Find $a : b : c$. **[4]**

23. Two numbers are in the ratio 5 : 8. Their sum is 78. Find the numbers. **[3]**

24. The ratio of the ages of A and B is 4 : 7. In 6 years the ratio will be 5 : 8. Find their present ages. **[5]**

25. A recipe uses flour, sugar and butter in the ratio 5 : 3 : 2. If 1.2 kg of flour is used, how much sugar and butter are needed? **[4]**

End of Worksheet • *Show all working clearly.*